

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Mechanical Engineering) SEMESTER - 6 Winter 2025 (Supply.)

Course :Bachelor of Technology (Mechanical Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 6

Subject Code & Name: BTMPE604A - PROCESS EQUIPMENT DESIGN

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question) 12

- 1 Non-destructive Testing is mainly used to: A. Improve Mechanical properties B. Detect internal and surface defects without damaging the component C. Increase production rate D. Reduce manufacturing cost
- 2 Ultrasonic testing works on the principle of: A. Reflection of sound waves B. Magnetic flux leakage C. Optical refraction D. Thermal radiation
- 3 Horizontal pressure vessels are commonly used for: A. Tall column applications B. Gas–liquid separation C. High vacuum service only D. Very high pressure storage
- 4 Which of the following is an atmospheric storage vessel? A. LPG sphere B. Gas cylinder C. Open water tank D. High-pressure reactor
- 5 Floating roof tanks are preferred mainly to: A. Increase tank strength B. Reduce vapor losses C. Increase operating pressure D. Improve aesthetics
- 6 The primary function of a gasket is to: A. Increase joint strength B. Prevent leakage between mating surfaces C. Reduce friction D. Absorb vibration
- 7 Which gasket is suitable for very high-pressure service? A. Rubber gasket B. PTFE gasket C. Ring type joint (RTJ) gasket D. Compressed fiber gasket
- 8 Which flange is used to close the end of a pipeline? A. Weld neck flange B. Slip-on flange C. Blind flange D. Lap joint flange
- 9 Autofrettage is a process used mainly to: A. Increase ductility of materials B. Induce residual compressive stresses in pressure vessels C. Reduce manufacturing cost D. Improve surface finish
- 10 An agitator is mainly used to: A. Increase pressure B. Mix fluids uniformly C. Reduce temperature D. Separate solids
- 11 Which agitator is suitable for high-viscosity fluids? A. Turbine agitator B. Propeller agitator C. Anchor agitator D. Jet mixer
- 12 Skirt support is commonly used for: A. Horizontal vessels B. Vertical pressure vessels C. Storage tanks only D. Heat exchangers only

Q2. Solve the following.

- A) Define pressure vessel. Explain in detail types of process equipment.

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- B) Write a short note on i] Liquid penetrating testing ii] Magnetic Particle testing ii] Ultrasonic Testing 6
- Q3. Solve the following.
- A) Explain Design method of atmospheric storage vessel. 6
- B) Write a short note on i] Storage of volatile liquids ii] Non volatile liquid 6
- Q4. Solve Any Two of the following.
- A) Explain unfired process vessel with internal and external pressure. 6
- B) Define Gaskets. Explain Different types of gasket. 6
- C) Define Blind Flange. Explain types of Blind flange. 6
- Q5. Solve Any Two of the following.
- A) What is thick cylinder ? Explain Autofrettage in details. 6
- B) Describe the materials for High pressure vessels. 6
- C) The piston rod of a hydraulic cylinder exerts an operating force of 10kN. The friction due to piston packing and stuffing box is equivalent to 10% of the operating force. The pressure in the cylinder is 10MPa. The cylinder is made of cast iron FG 200 and the factor of safety 5. Determine the diameter and the thickness of the cylinder. 6
- Q6. Solve Any Two of the following.
- A) Explain working and types of Agitated vessels with diagram. 6
- B) Why stresses occur at saddles. Describe types of stresses at saddle supports. 6
- C) Explain How power required for Agitation Estimate 6

*** End ***